Active ants: analysis

- $$\begin{cases} \mathrm{d}X_t^i & = \mathrm{Pe}v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i & = \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

- Well-posedness

- Locally Lipschitz, moment controlled ∇K and $\sigma_x \geq 0$: propagation of chaos on bounded time intervals (McKean, 1967; Bolley et al., 2011)

- Propagation of chaos: the empirical measure converges to the solution of the mean-field limit PDE as $N \to \infty$

- $\mathbb{E}\left[d(\mu_t^N, f_t)\right] \to 0$ as $N \to \infty$

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- Well-posedness
  - Mean-field limit: large stochastic system $\Longleftrightarrow$ deterministic PDE